



BRIDGING THE ENGINEERING-MEDICINE GAP

Enterprise Healthcare **Software & Biomedical** Solutions

Empowering hospitals, diagnostic imaging centers, and clinical laboratories across Ghana and West Africa with unified digital health platforms and precision medical devices.



📍 HIL Biomedicals Clinical Team on Site • Accra Medical Centre, Ghana

One Unified Partner for Software & Medical Devices

Bridging hospital clinical workflows, diagnostic laboratories, radiology, and physical medical equipment with single-source accountability.

Digital Health Systems

End-to-end clinical and hospital administrative automation designed for Ghanaian healthcare regulations and NHIS.

- Electronic Medical Records (EMR)
- Hospital Management System (HMS)
- Radiology PACS & Tele-imaging
- Integrated Diagnostic LIMS

Medical Equipment

Direct procurement, installation, and commissioning of international-standard medical hardware.

- Diagnostic Digital X-Ray & Ultrasound
- Critical Care ICU Ventilators & Monitors
- Operating Theater Lights & Tables
- Automated Biochemistry Analyzers

Integration & Maintenance

Seamless bidirectional device interfacing backed by certified biomedical calibration engineers.

- Direct HL7 & DICOM 3.0 Device Telemetry
- Planned Preventive Maintenance (PPM)
- Traceable Metrology & ISO Calibration
- Accra Bonded Spare Parts Inventory

The HIL Biomedicals Advantage

Unlike traditional vendors who supply software without understanding biomedical hardware—or hardware vendors who cannot integrate with your IT systems—HIL Biomedicals delivers a unified engineering ecosystem. Every patient monitor, laboratory analyzer, and X-ray machine connects directly into your clinical records with zero manual transcription errors.

99.9%

SYSTEM UPTIME

100%

HL7 / DICOM COMPLIANT

< 2 Hrs

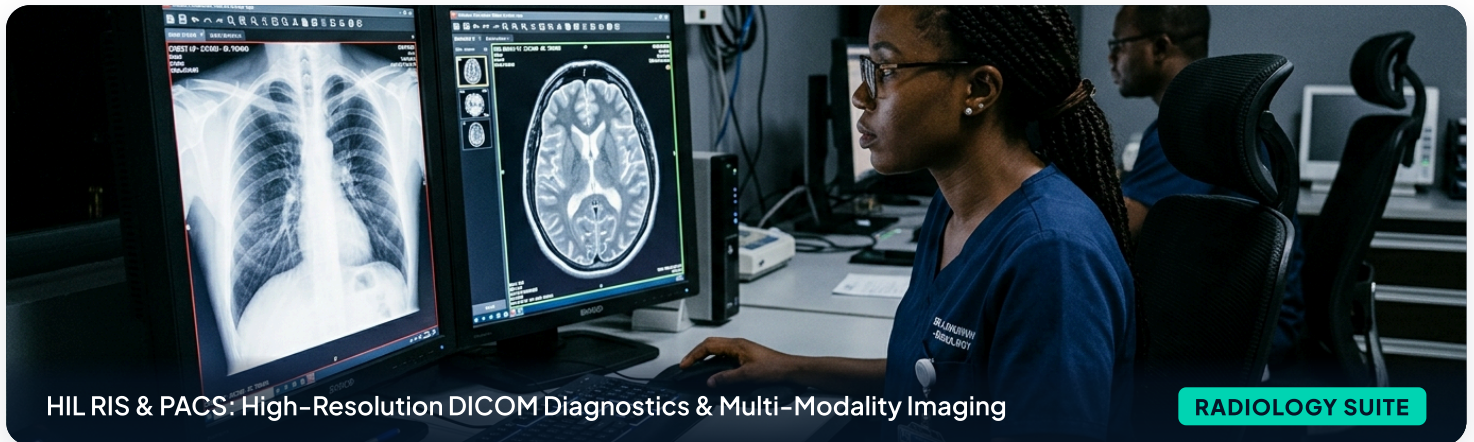
ACCRA RAPID RESPONSE

ISO 13485

CERTIFIED CALIBRATION

Clinical Systems: EMR, HMIS, RIS & PACS

Automating patient documentation, hospital administration, and multi-modality diagnostic radiology reporting.



HIL RIS & PACS: High-Resolution DICOM Diagnostics & Multi-Modality Imaging

RADIOLOGY SUITE

Electronic Medical Records

Longitudinal patient records eliminating lost physical folders and tracking clinical history.

- Custom doctor consultation templates
- Electronic prescriptions & vitals charting
- Bedside nurse medication administration
- Full audit logging and access controls

Hospital Management (HMIS)

Centralized patient registration, billing, ward admissions, and cashier reconciliation.

- OPD triage and appointment booking
- NHIS claims vetting and tariff rules
- Inpatient ward bed tracking and transfers
- Automated billing across all services

RIS & PACS Imaging

DICOM 3.0 compliant Picture Archiving & Communication for instant radiologist access.

- Direct capture from X-Ray, CT, MRI & Ultrasound
- Web-based diagnostic DICOM viewer
- Structured tele-radiology reporting
- Secure long-term cloud/local archiving

Diagnostics & Enterprise: LIMS, HFMS, SFMS & EMS

Laboratory analyzer interfacing, revenue leakage prevention, and multi-branch enterprise coordination.



HIL LIMS: Direct Bidirectional Interfacing with Automated Chemistry & Hematology Analyzers

**DIAGNOSTIC
LABORATORY**

Laboratory Information (LIMS)

Automates diagnostic testing workflows and sample accessioning.

- Barcode sample tracking & positive patient ID
- Direct ASTM/HL7 analyzer bidirectional data flow
- Automated doctor notification & signed PDF delivery

Financial Management (HFMS)

Guards hospital cashflow and eliminates billing revenue leakage.

- Departmental cashier audit trails and reconciliation
- Real-time NHIS tariff processing & claims tracker
- Procurement, general ledger & payroll integration

Smart Facility Management (SFMS)

Protects medical equipment investments and physical plant assets.

- Biomedical asset registry with maintenance histories
- Automated calibration and PPM schedule alerts
- Breakdown ticketing and spare parts inventory

Enterprise Medical Suite (EMS)

Consolidates multi-facility hospital chains and health networks.

- Centralized patient record across all branches
- Executive BI dashboards & bed occupancy analytics
- Ministry of Health & GHS regulatory reporting



Medical Equipment & Hardware Solutions

Turnkey clinical equipment provisioning, installation, and commissioning for modern hospital departments.

Diagnostic Medical Imaging

High-clarity radiological imaging equipment engineered for durability and low operating cost.

- **Digital Radiography (DR) X-Ray:** Floor-mounted & ceiling suspension systems with high-resolution Flat Panel Detectors (FPD).
- **Ultrasound Diagnostic Systems:** 3D/4D Color Doppler with cardiac, obstetric, and abdominal multi-frequency probes.
- **Surgical C-Arm & Mobile X-Ray:** Real-time intraoperative fluoroscopy for orthopedic and theater surgery.

Critical Care & ICU Life Support

Reliable physiological life support instrumentation for Intensive Care Units and emergency wards.

- **ICU Mechanical Ventilators:** Advanced invasive/non-invasive ventilation modes with integrated turbine drive.
- **Multi-Parameter Patient Monitors:** 12.1" touch screens tracking ECG, SpO2, NIBP, Respiration, Temperature & EtCO2.
- **Syringe & Infusion Pumps:** Micro-infusion volumetric pumps with anti-bolus safety alarms.

Clinical Diagnostic Laboratory

Automated clinical instrumentation with built-in digital interfaces for rapid throughput.

- **Automated Biochemistry Analyzers:** 200–400 tests/hour with refrigerated reagent compartments and barcode readers.
- **5-Part Differential Hematology Analyzers:** Comprehensive complete blood count (CBC) with laser flow cytometry.
- **Electrolyte & Blood Gas Analyzers:** Direct ISE measurement for emergency and critical care panels.

Surgical & Operating Theater

Surgical infrastructure designed for sterility, precision, and surgical team comfort.

- **LED Shadowless Operating Lights:** Dual-arm surgical lighting with cold light technology and intensity memory.
- **Electro-Hydraulic Operating Tables:** Multi-position surgical beds suitable for general, ortho, and neurosurgery.
- **Electrosurgical Units (Diathermy):** Monopolar and bipolar cutting/coagulation with contact monitoring.

Quality Assurance & Global Standards Compliance

Every medical device supplied by HIL Biomedicals is internationally certified (CE Marked, US FDA Listed, ISO 13485) and backed by local warranty, OEM spare parts, and on-site engineering support based in Accra, Ghana.

Hardware-to-Software Device Integration

Real-time physiological telemetry, automated analyzer data transfer, and elimination of manual transcription.

The HIL Interoperability Pipeline

Clinical data flows seamlessly from the physical patient bedside into the hospital electronic medical record via standardized protocol translation gateways.

1 Medical Device Layer

Patient monitors, ventilators, infusion pumps, blood analyzers, and digital X-ray machines generate clinical measurements.

- Bedside patient vital signs telemetry
- Biochemistry and hematology result arrays
- Radiological DICOM image datasets

2 HIL Integration Gateway

Standardized protocol translation engine operating locally on the hospital network.

- HL7 v2.x and FHIR protocol parser
- ASTM 1381/1394 laboratory interfacing
- DICOM C-STORE, C-FIND & Worklist integration

3 Hospital EMR / Database Core

Unified clinical database storing longitudinal patient encounters with cryptographic integrity.

- Automated real-time vitals plotting on patient chart
- Critical panic value alerts to doctor smartphones
- Automatic billing ledger creation upon test execution

4 Clinical & Administrative Touchpoints

Instant secure access across desktop workstations, tablets, and nursing monitors.

- Doctor tablets during ward rounds
- Central ICU nursing station multi-bed overview
- Hospital executive revenue and occupancy analytics

Clinical Impact of Device Interoperability

Eliminates transcription errors by 100%, slashes laboratory result turnaround time from hours to seconds, and ensures that every medical procedure performed is automatically billed without cash leakage.

Preventive Maintenance, Calibration & Field Support

Dedicated biomedical engineering technical division protecting hospital assets and patient safety.



HIL Biomedical Workshop: Electrical Safety Testing & Calibration on Patient Monitors & Ventilators

Planned Maintenance (PPM)

Proactive scheduled inspections preventing costly catastrophic equipment breakdowns.

- Electrical safety testing (IEC 60601-1)
- Filter, O-ring, and sensor replacements
- Mechanical lubrication and pneumatic tuning
- Comprehensive inspection logbooks

ISO Traceable Calibration

Metrology testing verifying diagnostic accuracy and treatment safety margins.

- Ventilator volume and pressure calibration
- Defibrillator discharge energy verification
- Patient monitor NIBP & ECG simulation
- Official compliance certificates for accreditation

24/7 Field Engineering Support

Rapid on-site emergency repair backed by an inventory of genuine spare parts.

- Bonded parts depot in Accra for zero delay
- Guaranteed rapid on-site arrival times
- Temporary loaner units for critical ICU care
- Hospital clinical engineering staff training

ISO 13485

QUALITY CERTIFIED

IEC 60601

ELECTRICAL SAFETY

24 / 7

ON-CALL SUPPORT

100%

ACCREDITATION READY



Partner With Ghana's Premier Healthcare Engineering Provider

Whether establishing a new healthcare facility, upgrading clinical EMR systems, or procuring certified medical equipment, HIL Biomedicals & Systems Ltd delivers proven engineering excellence.

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Serving Ghana & West Africa

SCAN WITH MOBILE CAMERA

Official Interactive Digital Portfolio

Scan this QR code to explore the live interactive portfolio, view system specifications, and inspect code on GitHub:



<https://ayehakuffo7-arch.github.io/hil-biomedicals-portfolio/>